

Matching the Target Is Not Enough: Surrogate Failure in Gradient-Based Targeted Instruction Selection

Anonymous authors
Paper under double-blind review

Abstract

Targeted instruction selection often assumes that subsets better aligned with a small query set will yield greater downstream utility. We test this sufficiency assumption under an operational gradient-based targeting pipeline. We introduce Directional Second-Moment Coreset (DSMC), a greedy construction that provides a directly measurable target geometry. On Llama-2-7B MMLU, DSMC outperforms first- and second-order round-robin baselines by 1.55 and 0.88 points, respectively; a pre-registered Llama-3.2-3B replication reverses these small observed differences (-0.18 and -0.31 points), while DSMC and Random remain nearly tied on both stacks. Thus DSMC’s method-level advantage does not transfer. In contrast, the central BBH counterexample does: DSMC attains the lowest operational target discrepancy in every draw yet underperforms Random in every draw on both model stacks. Among the frozen compared subsets, DSMC also retains the lowest discrepancy under an evaluation-matched target serialization, although finer rankings are serialization-sensitive. On the defining query items, the operational cross-entropy surrogate improves without corresponding chain-of-thought exact-match gains. Better alignment under the studied gradient-based targeting construction is therefore not sufficient for better downstream utility.

1 Introduction

Instruction tuning draws from increasingly large and heterogeneous pools, while a deployment often requires only a narrower set of capabilities. Targeted instruction selection addresses this mismatch by using a small query set to identify candidate examples that should be most useful for a target task. Recent methods operationalize usefulness through semantic retrieval, gradient similarity, influence approximations, target subspaces, or task-specific rewards (Xia et al., 2024; Kang et al., 2024; Wang et al., 2025; Min et al., 2026). Across these formulations lies an intuitive sufficiency assumption: if a selected subset matches the target more faithfully, training on it should improve target utility.

This paper directly tests that assumption. Doing so requires more than observing that one selector loses to another: a weak downstream result could simply mean that the selector failed to optimize the target representation it was meant to match. We therefore construct a deliberately measurable target-matching instrument. Given unit-normalized projected gradients u , we represent a set by its directional second moment,

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \quad (1)$$

Under the quadratic kernel $k_2(u, v) = (u^\top v)^2$, empirical maximum mean discrepancy (MMD) is exactly the squared Frobenius distance between directional second moments. Our method, Directional Second-Moment Coreset (DSMC), greedily optimizes this set-level discrepancy. It consequently lets us verify whether target geometry is actually optimized rather than inferring success from downstream performance.

Method gains are stack-dependent, while the central failure mode is not. On Llama-2-7B MMLU at the 5% budget, DSMC outperforms First-RR and Second-RR by 1.55 and 0.88

points, with wins in all ten target-direction cells. Repeating the pre-registered comparison on Llama-3.2-3B reverses these small observed differences (-0.18 and -0.31 points over five direction-averaged blocks), while DSMC and Random remain nearly tied on both stacks. We therefore treat DSMC as a diagnostic instrument rather than a generally superior selector.

In contrast, the BBH geometry–utility reversal transfers across both tested model stacks. Across three query/selection draws, DSMC attains the lowest operational second-moment discrepancy yet loses to Random in every draw on each stack. Llama-2-7B shows a -2.94 percentage-point DSMC–Random gap and Llama-3.2-3B a smaller -0.89 point gap. All BBH SFT means are below their corresponding no-SFT references; the experiment therefore tests the narrower ordering question of whether better target alignment implies better utility among equally trained subsets, not whether SFT improves the base model.

Finally, we locate the break more precisely. On the same 64 query items used to define the selection signal, target-aware methods reduce the wrapped final-answer cross-entropy computed by the targeting pipeline, while chain-of-thought exact match does not improve. Query-to-test shift therefore cannot by itself explain the result. A serialization diagnostic narrows the claim: the behavior of bare-context cross-entropy differs across model stacks, so we do not claim that cross-entropy is generically misaligned with chain-of-thought evaluation. The supported conclusion concerns the exact operational surrogate used by the selection pipeline.

Contributions.

- We turn an implicit sufficiency assumption in gradient-based targeted instruction selection into a controlled empirical question: does lower target discrepancy imply higher downstream utility?
- We introduce DSMC as a measurable intervention on target geometry. It produces a consistent MMLU advantage over RR baselines on Llama-2-7B, but this method-level advantage does not transfer to Llama-3.2-3B.
- We provide a cross-stack geometry–utility counterexample on BBH: DSMC attains the lowest operational D_2 in 3/3 draws yet underperforms Random in 3/3 draws on each of two model stacks.
- The DSMC–Random counterexample is robust to the tested evaluation-matched target serialization. A same-item diagnostic further shows that improvement in the operational query surrogate need not be accompanied by improvement in the task metric.
- We delimit the result through no-SFT and Random references, role-specific representation assumptions, serialization sensitivity, matched-format controls, and contamination audits. These diagnostics resolve an empirical counterexample chain but do not identify a unique causal mechanism.

2 Target Matching with Directional Second Moments

2.1 Targeted instruction selection

Let $\mathcal{C} = \{x_i\}_{i=1}^N$ be a candidate instruction pool and $\mathcal{Q} = \{q_j\}_{j=1}^M$ a small target query set. Given a budget K , the goal is to select $\mathcal{S} \subset \mathcal{C}$, $|\mathcal{S}| = K$, such that supervised fine-tuning on $\mathcal{S}$ improves held-out target performance.

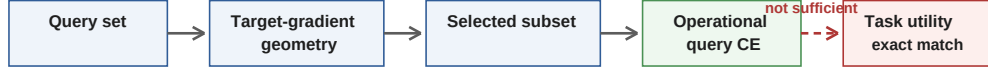
Following gradient-based targeted selection (Xia et al., 2024; Nayak et al., 2026), we compute per-example gradients at a shared warm-up checkpoint and project them to a reusable low-dimensional datastore. Following the optimizer-aware LESS protocol, candidates and queries use role-specific transforms before entering a common projected directional space:

$$u_{\mathcal{C}}(x) = \frac{\Pi A_{\text{Adam}} g(x)}{\|\Pi A_{\text{Adam}} g(x)\|_2}, \quad u_{\mathcal{Q}}(q) = \frac{\Pi g(q)}{\|\Pi g(q)\|_2}. \quad (2)$$

Here Π is a shared Rademacher projection and A_{Adam} denotes the candidate-side optimizer-aware transform. The MMD identity below is exact for these resulting unit vectors in the

(a) MMLU: the method gain is stack-dependent

	Llama-2-7B	Llama-3.2-3B	
DSMC - First-	+1.55 pp	-0.18 pp	→ gain present
DSMC - Secor	+0.88 pp	-0.31 pp	→ gain absent

(b) BBH: the surrogate chain breaks on both stacks**DSMC relative to no-SFT (three draw means)**

	Target D2	Wrapped CE	Same-item EM	Held-out EM
Llama-2-7B	0.086 < Random 0.110	-1.16 nats	-5.21 pp	-3.33 pp
Llama-3.2-3B	0.064 < Random 0.075	-3.25 nats	-6.51 pp	-0.97 pp

Geometry and the operational surrogate improve; the task metric does not.

Figure 1: Method gains are stack-dependent, while the BBH failure mode transfers. The MMLU comparison above the chain shows that DSMC’s advantage over RR baselines is present on Llama-2-7B but absent on Llama-3.2-3B. The BBH rows show the cross-stack counterexample: DSMC has lower D_2 than Random and improves the operational wrapped final-answer cross-entropy, while task exact match falls. Each D_2 box compares DSMC with Random; CE and EM boxes report DSMC changes relative to no-SFT. Arrows denote a tested surrogate chain, not an identified causal mechanism.

common projected space. The construction should not, however, be interpreted as applying one symmetric raw-gradient representation map to candidates and queries.

2.2 Why a directional second moment?

First-order mean-gradient matching represents a target by $\mathbb{E}[u]$. This signed mean can cancel when useful examples occupy opposite directions, even if they share the same underlying orientation. We instead use the non-centered directional second moment

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \quad (3)$$

Because $uu^\top = (-u)(-u)^\top$, this representation captures orientation rather than sign. We deliberately call it a directional second moment, not a covariance: the vectors are normalized and the mean is not subtracted.

Consider the positive semidefinite kernel

$$k_2(u, v) = (u^\top v)^2. \quad (4)$$

Since $\langle uu^\top, vv^\top \rangle_F = (u^\top v)^2$, the biased empirical MMD between a selected set $\mathcal{S}$ and query set $\mathcal{Q}$ is exactly

$$\begin{aligned} \text{MMD}_{k_2}^2(\mathcal{S}, \mathcal{Q}) &= \left\| \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} u_s u_s^\top - \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} u_q u_q^\top \right\|_F^2 \\ &\equiv D_2(\mathcal{S}, \mathcal{Q}). \end{aligned} \quad (5)$$

The identity is algebraic: DSMC matches this empirical second moment exactly as defined, but the kernel is not characteristic and does not identify the full gradient distribution.

2.3 Exact greedy cores set construction

For a candidate x , define target relevance and accumulated selected-set similarity as

$$r_{\mathcal{Q}}(x) = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} k_2(u_x, u_q), \quad r_{\mathcal{S}}(x) = \sum_{s \in \mathcal{S}} k_2(u_x, u_s). \quad (6)$$

At step $m = |\mathcal{S}|$, the exact one-step marginal for minimizing the biased MMD selects

$$x^* = \arg \max_{x \notin \mathcal{S}} \left[r_{\mathcal{Q}}(x) - \frac{r_{\mathcal{S}}(x) + k_2(u_x, u_x)/2}{m+1} \right]. \quad (7)$$

For unit vectors $k_2(u_x, u_x) = 1$. The first term rewards query alignment and the second penalizes selected-set self-similarity. We use the deterministic exact marginal throughout; we make no stochastic-greedy or submodularity guarantee.

Role in this paper. DSMC is not claimed to optimize downstream utility. Its purpose is to construct a strong, measurable intervention on target geometry. If DSMC lowers D_2 but utility fails to improve, the discrepancy cannot be attributed to the geometry objective being left unoptimized.

3 Experimental Design

3.1 Candidate pool, features, and training

The candidate pool contains 270,679 processed examples from Flan v2 (100,000), CoT (100,000), Dolly (15,011), and OASST1 (55,668), matching the pool used by LESS-style targeted selection. For each model stack, a LoRA warm-up checkpoint is used to construct a candidate-gradient datastore. We use an 8,192-dimensional Rademacher projection with seed 123, unit-normalize the projected vectors, and generate query-gradient caches separately for each draw. The selected indices are materialized into static subsets; all downstream comparisons restart from the corresponding base model rather than continuing the warm-up run.

Downstream SFT uses LoRA rank 128, alpha 512, dropout 0.05 on q/k/v/o_proj, learning rate 2×10^{-5} , linear decay, 3% warm-up, bfloat16, effective batch size 128, and four epochs. The 5% budget is $K = 13,533$ (420 optimizer steps) and the 1% budget is $K = 2,707$ (84 steps). Full configurations, prompt templates, hashes, and data-processing details are provided in Appendix H.

3.2 Target families

MMLU controlled stress test. We construct ten globally disjoint 64-example query sets: five STEM-majority draws (51 STEM, 13 humanities) and five humanities-majority draws with the proportions reversed. Evaluation is balanced accuracy over held-out MMLU STEM and humanities subjects (Hendrycks et al., 2020). Draw index is paired with SFT seed $\{42, 1, 2, 3, 4\}$; for comparisons shared across the two skew directions, the primary descriptive unit is the five direction-averaged draw-index blocks.

BBH external validation. We stratify 27 BIG-Bench Hard tasks into a query reservoir and a frozen held-out split (Suzgun et al., 2022). Three 64-example query draws define the selection signals; the held-out evaluation contains 5,209 examples and uses pinned chain-of-thought few-shot exact match. Each query/selection draw is crossed with two SFT seeds, which are averaged within the draw. The statistical unit is therefore the draw ($n = 3$), not the six seed-level runs.

3.3 Comparators and reporting policy

The controlled gradient baselines are: first-order mean-gradient TopK (LESS-style), first- and second-order relevance TopK, first- and second-order round-robin (RR), GIST whitening

Table 1: Llama-2 MMLU representation $\times$ selector attribution. Balanced accuracy at one fixed seed for each skew direction. In this stack, the second-order representation improves both TopK and MMD; the smaller DSMC–Second-TopK increment is within the measured training-noise scale and is not treated as a stand-alone statistical claim.

Target	1st-TopK	1st-MMD	2nd-TopK	DSMC
STEM-majority	39.77	38.47	40.68	41.10
Humanities-majority	39.31	38.24	39.92	40.54

Table 2: Llama-2 MMLU target-draw results. Mean balanced accuracy over ten query draws (five paired draw-index blocks). In this stack, DSMC is the strongest mean targeted method at both budgets, but it shows no observed advantage over Random at 5% and trails Random at 1%. The no-SFT reference is shared, not a replicate.

Method	5%	1%
No-SFT	40.03	40.03
LESS-style TopK	39.41	39.37
First-RR	38.80	38.20
Second-RR	39.48	40.00
GIST-SharedProj	38.85	39.45
NICE-MMLU-EM	39.25	39.42
DSMC	40.36	40.17
Random- K	40.37	40.95
Length-matched Random	40.42	40.80

on the shared projected datastore (GIST-SharedProj), and a task-metric adaptation of NICE (NICE-MMLU-EM). Because several are shared-protocol adaptations rather than full official end-to-end reproductions, we use these qualified names in the paper. Target-independent controls are uniform Random- K , length-matched Random on MMLU, a sequence/label-length-matched Random control on BBH, and a shared no-SFT reference.

The study is descriptive. We report draw-level means, paired differences, and win counts; we do not report significance tests over $n = 3$ or $n = 5$ blocks. Subtask breakdowns and individual SFT seeds are diagnostics, not independent replicates.

4 Method Gains Are Stack-Dependent on MMLU

4.1 Llama-2 positive control

We first ask whether the proposed instrument is non-vacuous. Table 1 factors the method into target representation (signed first order versus directional second order) and selection rule (relevance TopK versus MMD coreset). Moving to the second-order representation improves both selectors in both skew directions: +0.61–0.91 points under TopK and +2.30–2.63 points under MMD. The selector interaction is representation dependent: first-order MMD hurts relative to first-order TopK, while second-order MMD adds 0.42–0.62 points over second-order TopK. The latter comparison is single-seed and small, so this is a Llama-2 positive control, not a cross-stack method claim.

Across ten target draws at the 5% budget, DSMC beats LESS-style TopK and both RR baselines in every draw, and beats the examined GIST and NICE adaptations in nine of ten draws (Table 2). Relative to Second-RR, the mean gain is 0.88 points and holds in all ten cells, supporting a complementary coreset effect at this budget. The positive claim is deliberately scoped: in the studied Llama-2 MMLU setting, directional second moments materially improve target-aware selection.

The absolute comparison creates the first reversal. Against Random- K , DSMC has a -0.01 point mean difference at 5%, winning two of five direction-averaged blocks; length-matched Random gives the same qualitative conclusion. Tightening the budget does not rescue target

Table 3: Cross-stack MMLU comparison at the 5% budget. DSMC’s advantage over the two RR baselines is consistent on Llama-2-7B but does not transfer to Llama-3.2-3B. DSMC and Random have nearly tied observed means on both stacks. Win counts use the pre-specified descriptive units for each experiment; no significance test is claimed.

Stack	DSMC – First-RR	DSMC – Second-RR	DSMC – Random
Llama-2-7B	+1.55 (10/10)	+0.88 (10/10)	−0.01 (4/10)
Llama-3.2-3B	−0.18 (2/5)	−0.31 (1/5)	+0.12 (3/5)

Table 4: Held-out BBH exact match. Means first average the two SFT seeds within each of three query/selection draws. All method means are below the corresponding no-SFT reference. DSMC loses to Random in every draw on both model stacks, while achieving the lowest target second-moment discrepancy among the compared subsets in every draw.

Stack	Method	Draw 0	Draw 1	Draw 2	Mean	Δ vs. no-SFT
Llama-2-7B	No-SFT	39.64				–
	Random- K	39.30	38.78	39.67	39.25	−0.39
	First-RR	36.24	37.61	37.61	37.15	−2.49
	Second-RR	36.80	37.00	36.80	36.87	−2.77
	DSMC	36.36	36.47	36.10	36.31	−3.33
Llama-3.2-3B	No-SFT	47.11				–
	Random- K	47.32	46.61	47.17	47.03	−0.08
	First-RR	46.11	46.52	47.04	46.56	−0.55
	Second-RR	46.62	46.23	47.18	46.68	−0.44
	DSMC	45.98	45.88	46.57	46.14	−0.97

awareness. At 1%, the DSMC–Random gap becomes −0.77 points and no query-direction cell favours DSMC. The extra MMD gain over Second-RR also contracts to 0.17 points and five of ten cells.

The no-SFT line changes the interpretation of the targeted leaderboard. Most targeted methods fall below the base reference at both budgets. DSMC stays slightly above it, but Random is the strongest absolute method at 1%. An equal-step sensitivity repeats the 1% subsets to 420 steps; both DSMC and Random fall to approximately 38.3%, so extra optimization does not rescue the targeted subset (Appendix E).

4.2 Pre-registered cross-stack check

The pre-registered Llama-3.2 comparison gives the result least favourable to a general method claim. Mean balanced accuracy is 47.95 for Second-RR, 47.82 for First-RR, 47.64 for DSMC, and 47.52 for Random, while the no-SFT reference is 49.47. All four methods lie within 0.44 points and below base. Table 3 shows the sharper contrast: DSMC’s Llama-2 advantage over both RR arms reverses on Llama-3.2, whereas its difference from Random remains close to zero on both stacks. Thus the method-level advantage of DSMC does not transfer. In contrast, the BBH geometry–utility counterexample below does.

5 External Reversal on BBH

MMLU queries are deliberately skewed, leaving open the possibility that the selector faithfully follows an unrepresentative query sample. BBH removes this specific explanation: queries and evaluation examples are held-out samples from the same task family. The result is stronger, not weaker (Table 4). On Llama-2-7B, Random reaches 39.25% micro exact match, while DSMC reaches 36.31%, a −2.94 point paired gap with zero of three draws favouring DSMC. Both RR variants also outperform DSMC; together with Table 3, this shows that a favourable DSMC ranking is neither cross-task nor cross-stack.

The direction repeats on Llama-3.2-3B. Random remains best at 47.03%, and DSMC is last among the four shared arms at 46.14%. The -0.89 point gap is negative in all three draws. The magnitude is substantially attenuated: all methods are within one point of the no-SFT model, so the two stacks should not be presented as equally dramatic. They nevertheless agree on the load-bearing directional fact.

All SFT means are below no-SFT. On Llama-2, Random is only 0.39 points below the shared base reference, while target-aware methods are 2.49–3.63 points below it. On Llama-3.2, Random is 0.08 points below base and DSMC 0.97 below. These are shared reference lines rather than replicated treatments, so we describe the observed means rather than claim inferentially established negative transfer.

Matched-format control. On Llama-2, constraining Random to match DSMC’s coarse sequence-length $\times$ loss-bearing-label-length histogram lowers Random from 39.25% to 38.05%, but still leaves it 1.74 points above DSMC. This is consistent with instruction-format or correlated provenance composition contributing to the gap. It is not a causal decomposition: the control changes several correlated properties simultaneously. Full results are in Appendix E.

6 Where the Surrogate Chain Breaks

6.1 The target geometry is optimized

For every BBH draw, we compute the operational discrepancy greedily optimized by DSMC,

$$D_2(\mathcal{S}, \mathcal{Q}_d) = \|M_{\mathcal{S}} - M_{\mathcal{Q}_d}\|_F^2. \quad (8)$$

On Llama-2, mean D_2 is 0.0863 for DSMC and 0.1098 for Random; on Llama-3.2 it is 0.0644 and 0.0752. More importantly, DSMC has the lowest D_2 among the compared subsets in all three draws of both stacks, while being worse than Random downstream in all three draws. This direct 3/3-versus-3/3 counterexample is the primary geometry result; descriptive rank correlations are reported only in Appendix C.

This rules out the narrow explanation that DSMC loses because its objective was not actually optimized. In MMLU, DSMC is also closer than Random to a leave-one-draw-out balanced reference in all ten draws at both budgets (Appendix C). Thus the observed failure is not simply that DSMC follows a skewed query away from the balanced reference.

Evaluation-matched target serialization. To test whether the DSMC minimum is induced by the chat serialization used for target-gradient extraction, we re-extract only the target gradients under the evaluation-matched bare serialization and rescore the frozen subsets. The change is nontrivial: mean row cosine between operational and evaluation-matched target gradients is only 0.25–0.36. Among the frozen compared subsets, DSMC retains the lowest evaluation-matched D_2 in all three draws on both stacks, while remaining worse than Random downstream in all three draws. The finer ranking is not invariant: on Llama-3.2, the four-method geometry ordering changes in every draw and Random moves from last to second. We therefore claim robustness of the DSMC–Random counterexample to this tested target-serialization change, not serialization-invariant target-space rankings. Full values appear in Appendix D.

6.2 The operational query surrogate improves

Gradient-based targeting uses a differentiable final-answer objective to construct query gradients. We therefore recompute teacher-forced final-answer cross-entropy on the same query items, under the exact wrapped serialization used by each stack’s targeting pipeline. On Llama-2, all four target-aware methods reduce this loss by 0.68–1.40 nats, while Random increases it by 0.33. On Llama-3.2, the target-aware methods reduce it by 3.19–3.40 nats and Random by only 1.37. The second stack therefore gives a magnitude split rather than a sign split; in both, the targeted methods move the operational surrogate more.

Table 5: The BBH surrogate chain for DSMC relative to no-SFT. Δ CE and same-item exact match use the 64 query items; held-out exact match uses 5,209 examples. “ D_2 wins” counts draws in which DSMC has the lowest discrepancy among the compared subsets; CE and EM entries are changes relative to no-SFT.

Stack	D_2 wins	Δ wrapped CE	Δ query EM	Δ held-out EM
Llama-2-7B	3/3	−1.159	−5.21	−3.33
Llama-3.2-3B	3/3	−3.255	−6.51	−0.97

6.3 Task utility does not follow on the same examples

Comparing query loss to held-out exact match changes both metric and examples. We consequently evaluate chain-of-thought exact match on the same 64 query items. For Llama-2, every target-aware method’s mean query exact match falls relative to base (−3.65 to −6.51 points) despite the substantial cross-entropy reductions. On Llama-3.2, all three targeted methods again have negative mean changes, with DSMC falling by 6.51 points; Random changes by +1.30 points. The individual 64-item draw metrics are coarse—one item is 1.56 points—but the mean separation is consistent with the held-out ranking.

The same-item result matters logically. A pure account in which targeted selection improves the query examples but fails to generalize cannot explain the observation, because the task metric does not improve on the query examples themselves. It does not rule out every form of overfitting, nor does it identify the missing mechanism.

6.4 Serialization narrows the surrogate claim

The operational query loss uses each stack’s LlamaFactory wrapper, whereas BBH generation uses a pinned bare evaluation context. Recomputing final-answer cross-entropy on that bare context yields different behavior across stacks. On Llama-2, no method improves bare-context CE, although targeted methods degrade it much less than Random. On Llama-3.2, target-aware methods do improve bare-context CE while Random does not. Therefore we do not claim that final-answer cross-entropy is inherently misaligned with chain-of-thought generation. The claim is narrower and operational: the surrogate actually used by the targeting pipeline can improve on the defining examples without a corresponding improvement in the task metric, and the role of serialization is model-stack dependent.

7 Related Work

Instruction data selection. General-purpose instruction selection commonly scores data for quality, difficulty, diversity, or representativeness. DEITA combines complexity and diversity for data-efficient alignment, while principled warm-up selection studies how small subsets affect later fine-tuning (Liu et al., 2024; Kang et al., 2024). Our focus is narrower: selection for a specific target described by a small query set.

Targeted and influence-based selection. LESS builds reusable low-dimensional gradient datastores and uses optimizer-aware gradient similarity for targeted instruction tuning (Xia et al., 2024). NICE replaces next-token-loss influence with a policy-gradient construction tied to non-differentiable evaluation metrics (Wang et al., 2025). GIST recovers a target-gradient subspace and scores candidate alignment under coupled optimization geometry (Min et al., 2026). GIO uses examples representing a target distribution to optimize a scalable gradient-information objective for dataset selection (Everaert & Potts, 2023). A recent controlled study separates representations from selection algorithms and finds that gradient representations are often predictive, but rankings vary across tasks, models, and budgets (Nayak et al., 2026). We do not compete with that study on breadth. We instead optimize a measurable set-level geometry, verify that the intended geometry is attained, and test whether that success is sufficient for utility.

Surrogate mismatch and strong Random baselines. ROSE motivates reward-oriented selection by documenting non-monotonicity between instruction-tuning loss and task reward (Wu et al., 2025); NICE makes a related move from differentiable training loss to task metrics (Wang et al., 2025). Large-scale studies also find that sophisticated selectors can be difficult to distinguish from or can underperform Random (Xia et al., 2025). Our contribution is not another broad selector leaderboard and not a replacement objective. It is a diagnostically resolved counterexample chain: target geometry is measurably optimized, the operational query surrogate moves in the intended direction, and task utility still does not follow.

8 Limitations

The study covers two model stacks (Llama-2-7B and Llama-3.2-3B), one candidate pool, and two target families. A model stack changes model, tokenizer, and serialization together; it is not an architecture-only ablation. BBH has only three independent query/selection draws, and the effect on the 3B stack is small. The MMLU method differences on Llama-3.2 are also small: all four methods are within 0.44 points and below the shared no-SFT reference. MMLU couples draw index to SFT seed, so direction-averaged blocks rather than ten cells carry comparisons shared across directions.

Candidates and queries use role-specific gradient transforms: candidate features are Adam-aware, whereas query features use raw SGD gradients. Our claims therefore concern the resulting operational push-forward geometry, rather than symmetric raw-gradient distribution matching in general. The evaluation-matched sensitivity changes target serialization but does not remove this asymmetry or answer how an SGD/SGD construction would behave. Moreover, although the frozen DSMC subsets remain closest under the tested bare serialization, the finer Llama-3.2 geometry ranking changes in every draw.

Several baselines are controlled adaptations on a common datastore, not full official end-to-end reproductions. The contamination audit includes exact, fuzzy, and long n-gram lexical checks but not a semantic nearest-neighbor audit. Finally, the diagnostics do not identify a unique mechanism. Format and provenance are plausible contributors, while a simple task-frequency specialization account is unsupported, but neither conclusion is causal.

9 Conclusion

What transfers across the two tested model stacks is not DSMC’s method advantage, but a counterexample to the sufficiency of the studied surrogate. In these settings, lower second-moment discrepancy under the operational gradient-based targeting construction does not guarantee higher downstream utility. Tighter optimization of a selection surrogate should therefore be treated as progress only after the surrogate’s relationship to the downstream task metric has been independently validated.

AI use statement

Generative AI tools were used to summarize and cross-check existing literature, inspect repository artifacts, suggest paper structure and wording, and draft and edit parts of the manuscript. Earlier in the research workflow, generative AI tools were also used to provide feedback on experimental design, interpretation, and code. The authors reviewed all AI-assisted text and claims against the underlying code, configurations, manifests, and result files; all reported numerical results were taken from experiment artifacts rather than generated by an AI system. The authors take responsibility for the final content, including all AI-assisted material.

Ethics statement

This work studies data selection for publicly available language-model benchmarks and does not involve human-subject data collection. The main risks are those shared by instruction tuning, including inherited dataset biases, benchmark contamination, and overgeneralization

from a limited experimental scope. We report negative results, adapted-baseline scope, and contamination audit limits explicitly to reduce the risk of overstating practical conclusions.

Reproducibility statement

Appendix H documents the data composition, target-draw construction, feature protocol, selection rules, training hyperparameters, evaluation pins, statistical units, and integrity checks. The supplementary material will include anonymized code, exact run configurations, frozen manifests and hashes, paper tables generated from committed result files, and the pre-registered protocols for the BBH and second-model experiments.

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A Method Derivations

A.1 MMD identity

Let $\widehat{M}_S = |\mathcal{S}|^{-1} \sum_{s \in \mathcal{S}} u_s u_s^\top$ and define $\widehat{M}_Q$ analogously. Expanding the squared Frobenius norm gives

$$\begin{aligned} \|\widehat{M}_S - \widehat{M}_Q\|_F^2 &= \frac{1}{|\mathcal{S}|^2} \sum_{s, s'} \langle u_s u_s^\top, u_{s'} u_{s'}^\top \rangle_F \\ &\quad + \frac{1}{|\mathcal{Q}|^2} \sum_{q, q'} \langle u_q u_q^\top, u_{q'} u_{q'}^\top \rangle_F \\ &\quad - \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s, q} \langle u_s u_s^\top, u_q u_q^\top \rangle_F \end{aligned} \quad (9)$$

$$= \frac{1}{|\mathcal{S}|^2} \sum_{s, s'} (u_s^\top u_{s'})^2 + \frac{1}{|\mathcal{Q}|^2} \sum_{q, q'} (u_q^\top u_{q'})^2 - \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s, q} (u_s^\top u_q)^2. \quad (10)$$

This is the biased empirical MMD under k_2 .

A.2 Greedy marginal

Terms depending only on $\mathcal{Q}$ can be dropped. Comparing the objective after adding x to a set of size m yields Equation 7 after positive rescaling. The implementation maintains $r_S(x)$ incrementally and masks selected or invalid rows. Zero and non-finite feature rows are rejected before selection.

B Additional MMLU Results

Pre-registered Llama-3.2 outcome. The 5% cross-stack arm completed 35/35 adapters and 36/36 evaluations with no failures. The statistical unit is five direction-averaged draw-index blocks. The no-SFT balanced reference is 49.47; mean balanced accuracy is 47.95 for Second-RR, 47.82 for First-RR, 47.64 for DSMC, and 47.52 for Random. DSMC minus First-RR is -0.18 points (2/5 blocks), minus Second-RR is -0.31 (1/5), and minus Random is $+0.12$ (3/5). This is the pre-registered outcome in which the Llama-2 method advantage does not transfer.

Lower budget and equal-step sensitivity. On Llama-2, DSMC minus Random changes from -0.01 points at 5% to -0.77 at 1%, with 0/10 1% cells favouring DSMC. Repeating the 1% DSMC and Random subsets to the 420 steps used by the 5% arm reduces both to approximately 38.3%, about 1.8 points below the shared no-SFT reference. Thus the low-budget comparison is not rescued by simply increasing optimizer steps.

C Additional Geometry Diagnostics

On MMLU, a leave-one-draw-out balanced reference gives mean D_2 values of 0.1538 for DSMC and 0.1762 for Random at 5%, and 0.1522 versus 0.1764 at 1%; DSMC is closer in all ten draws at each budget. On BBH, the descriptive Spearman correlation between D_2 and accuracy is positive in all three Llama-2 draws (0.77, 0.83, 0.89 over six methods) and all three Llama-3.2 draws (0.80, 1.00, 0.20 over four methods). These ranking samples are small and no inferential interpretation is attached to them.

D Evaluation-Matched Target-Geometry Sensitivity

This post-hoc check changes only target serialization; it does not rerun selection or SFT. Mean row cosine between operational and evaluation-matched target gradients is 0.335/0.342/0.363 for Llama-2 and 0.249/0.270/0.272 for Llama-3.2.

Table 6: Evaluation-matched D_2 for frozen BBH subsets. Lower is better. The frozen DSMC subset is closest in all six stack-draw cells, but the non-DSMC ranking changes in all three Llama-3.2 draws.

Stack	Draw	DSMC	First-RR	Second-RR	Random
Llama-2	0	0.06675	0.06830	0.07102	0.08875
	1	0.08230	0.08393	0.08586	0.10358
	2	0.05655	0.05800	0.05989	0.07831
Llama-3.2	0	0.07985	0.09134	0.08921	0.08847
	1	0.07660	0.08809	0.08619	0.08598
	2	0.07124	0.08414	0.08203	0.08051

E Sensitivity and Falsification Checks

Matched Random controls. MMLU length-matched Random gives the same qualitative target-awareness result as plain Random. On Llama-2 BBH, sequence×label-length-matched Random scores 38.05, between plain Random (39.25) and DSMC (36.31). The control changes correlated provenance and content as well as format, so it does not support a causal percentage decomposition.

Same-item and bare-context diagnostics. On Llama-2, targeted methods reduce operational wrapped query CE by 0.68–1.40 nats but their same-item CoT exact match changes by −3.65 to −6.51 points. Under the bare evaluation context, no method improves CE; targeted methods degrade it less than Random. On Llama-3.2, targeted methods reduce wrapped CE by 3.19–3.40 nats and also reduce bare CE by 0.41–0.57, while their mean same-item CoT changes remain negative. This is why the main claim is tied to the operational surrogate and the tested pipeline, not cross-entropy in general.

Task exposure. Per-task query frequency has negative descriptive correlations with targeted-vs-Random subtask deltas (−0.210, −0.280, −0.224 across the three draws). We find no evidence for a simple account in which greater query exposure protects a BBH task. This diagnostic is exploratory and does not identify an alternative mechanism.

F Baseline Implementations

LESS-style TopK uses one shared checkpoint and mean signed gradient similarity; it is not the official multi-checkpoint LESS pipeline. GIST-SharedProj applies the paper’s target-subspace whitening construction to the experiment’s shared normalized projected caches. NICE-MMLU-EM uses eight Monte Carlo samples and an exact-match task reward in the common feature protocol. These adaptations are intended to isolate representation and selection effects under controlled features; the paper does not claim exact end-to-end reproduction of the official systems.

G Contamination and Prompt Audits

Against MMLU and held-out BBH, normalized exact matching finds zero candidate duplicates. Full-pool fuzzy screening finds zero pairs above Jaccard 0.3; seven MMLU and five BBH 13-gram hits were manually reviewed and judged boilerplate false positives. The BBH few-shot audit finds no exact duplicates and is recorded as pass-with-disclosure for fuzzy near-duplicates. Prompt parity, few-shot leakage, and token-truncation audits are included in the supplementary artifact. A semantic embedding nearest-neighbor contamination audit was not performed, so these checks are lexical screens rather than a complete decontamination certificate.

H Reproducibility Details

The candidate pool contains 270,679 examples. Projected gradients use dimension 8,192 and seed 123. MMLU target draws are globally disjoint; BBH uses three stratified 64-example query draws and a 5,209-example held-out set. Frozen manifests record target files, gradient tensors, selections, materialized subsets, adapters, and evaluation-result hashes. All BBH evaluations pass 27-task/5,209-example gates, and all Llama-3.2 MMLU evaluations pass 57-task gates. Selection-level stability was checked after target-gradient non-bit-reproducibility was observed; the worst selector changed 5 of 2,707 indices.

H.1 Compute and selection overhead

Table 7: Representative measured or launch-accounted costs on eight NVIDIA H20 GPUs. Candidate datastores are reusable across target draws and tasks for a fixed model stack. Times for target gradients and selectors are coarse stage-level accounting rather than microbenchmarks.

Stage	Reuse scope	Representative cost
Warm-up checkpoint	per model stack	~1.5 GPU-h
Candidate gradient datastore	across targets	~6–8 GPU-h per stack
Three BBH target caches	per query draws	~1 GPU-h
DSMC + First/Second-RR	per three draws	~1.5 GPU-h
Random- K sampling	per subset	negligible
Llama-3.2 BBH SFT, 24 adapters	per selected subsets	3.39 GPU-h measured
Llama-3.2 MMLU SFT, 35 adapters	per selected subsets	16.88 GPU-h measured
Llama-2 BBH SFT, 36 adapters	per selected subsets	13.18 GPU-h measured

The main practical asymmetry is therefore not the greedy selector alone but the model-in-the-loop feature pipeline: Random requires neither warm-up nor gradient datastores. The anonymous supplement will include sanitized configurations and manifests. Before release, absolute cluster paths, Git metadata, user names, and repository links must be removed.